Place UZ Logo from Library

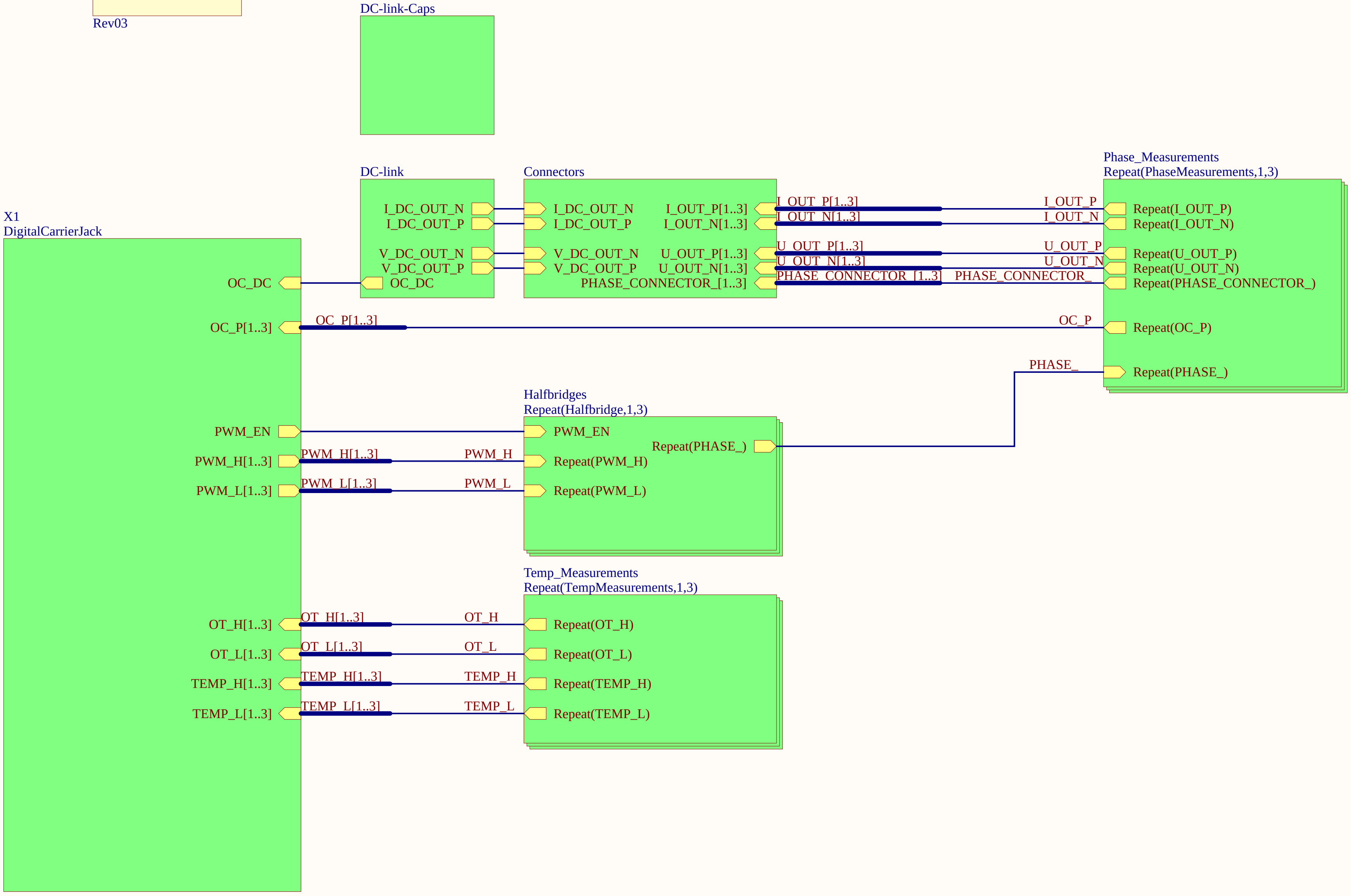
Revision

Rev03

Serial

Serial

Serial #



INFO

Project
ProjectRevision
AuthorParam
ProjectDate

Design Information

LOGO1

UltraZohm

UZ Logo

Title TopSheet.SchDoc	
Revision: Rev03	Design Engineer: D. Hufnagel
Project: UZ_D_Inverter.PrjPCB	

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Date: 17.04.2023

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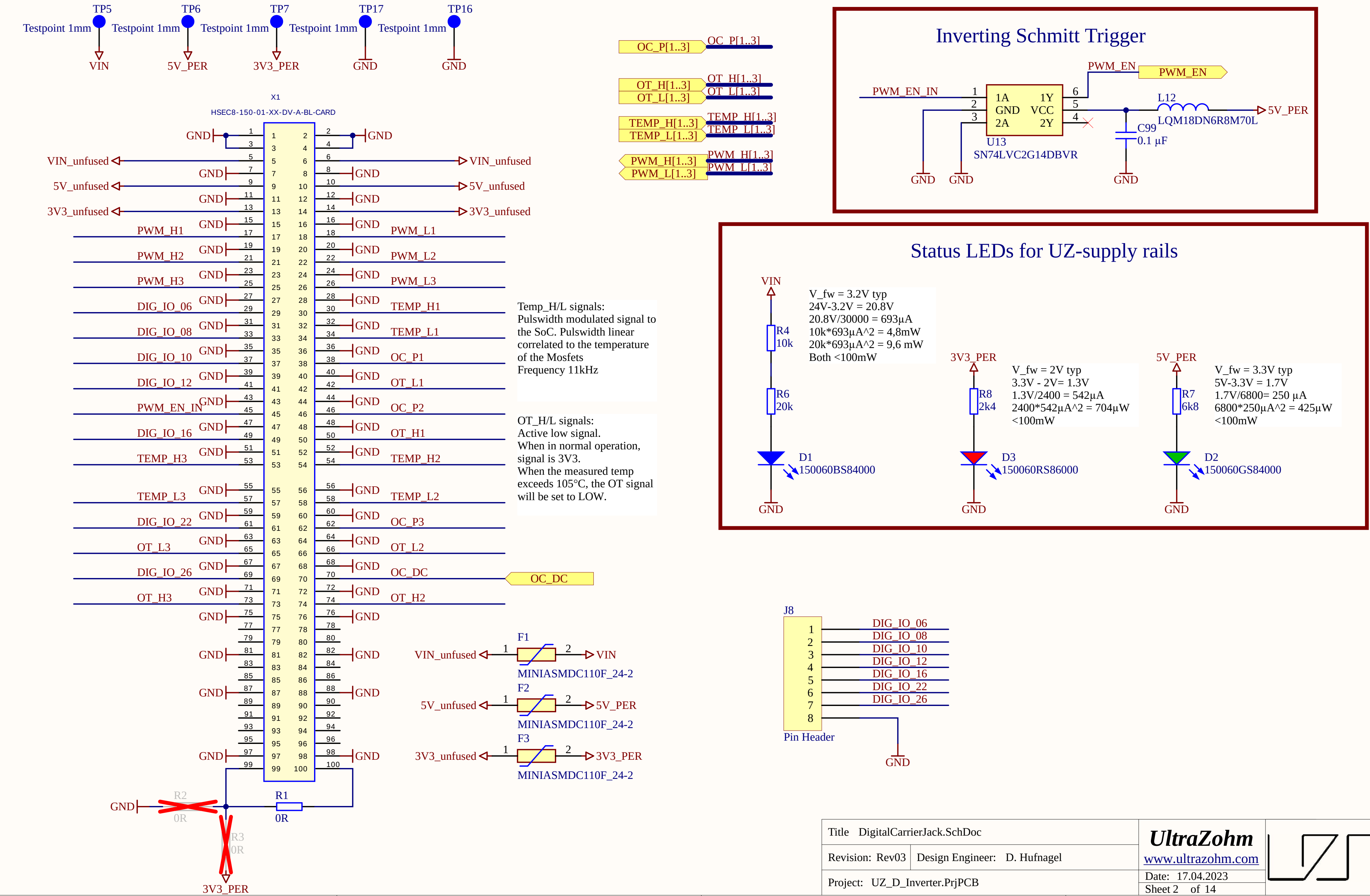
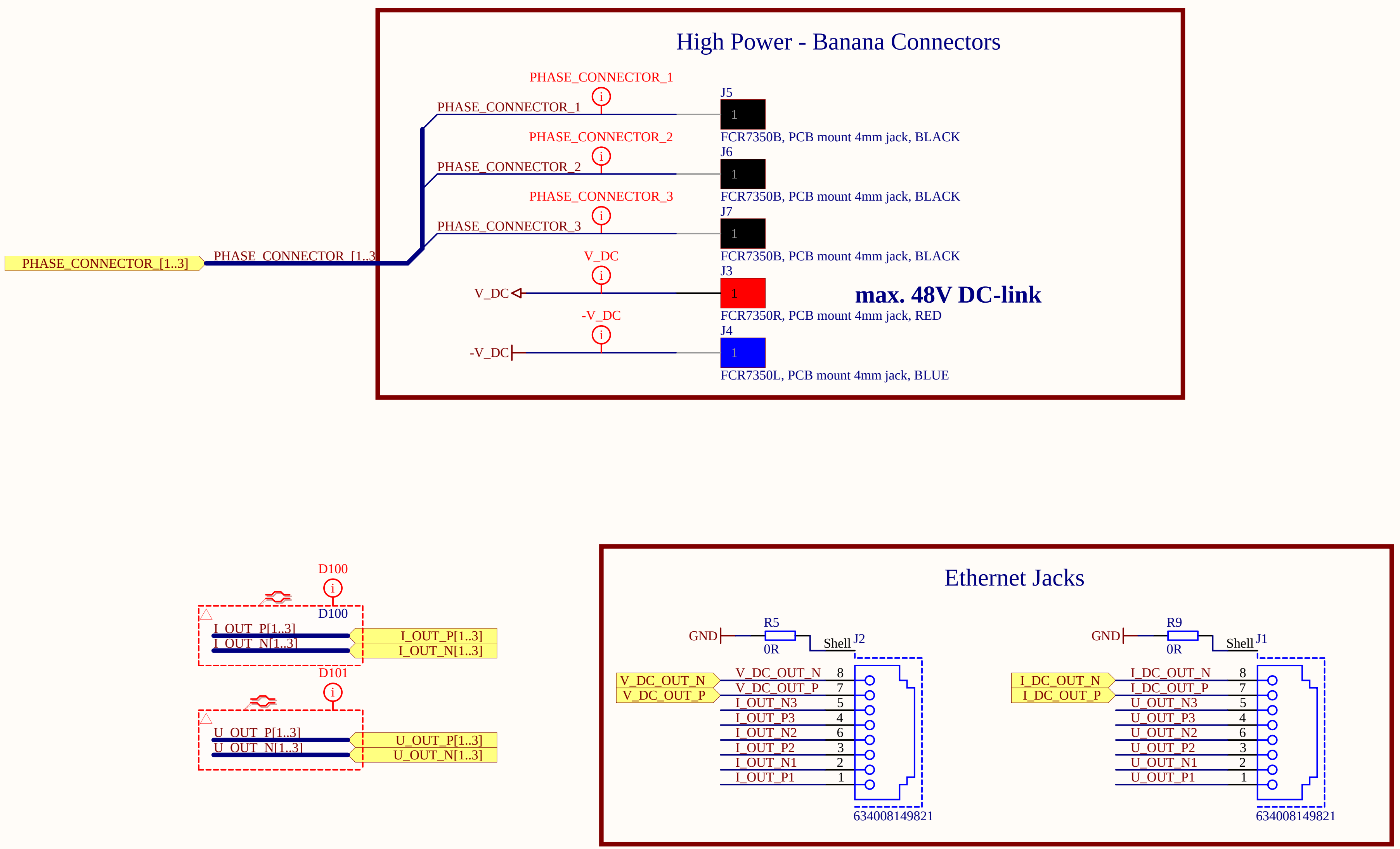
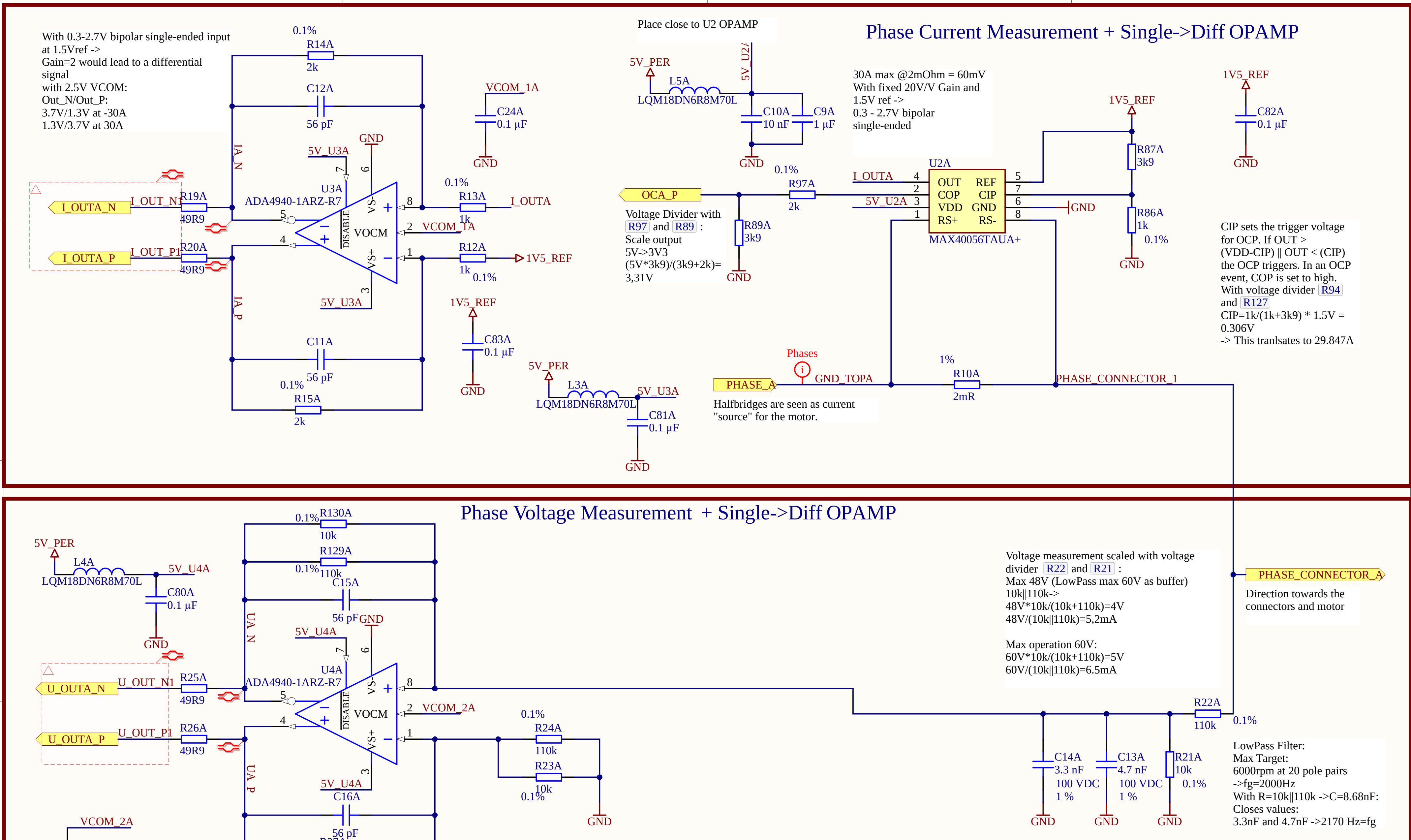


Title DigitalCarrierJack.SchDoc	*UltraZohm* www.ultrazohm.com
Revision: Rev03 Design Engineer: D. Hufnagel	
Project: UZ_D_Inverter.PrjPCB	Date: 17.04.2023 Sheet 2 of 14



Title Connectors.SchDoc		***UltraZohm*** www.ultrazohm.com
Revision: Rev03	Design Engineer: D. Hufnagel	Date: 17.04.2023
Project: UZ_D_Inverter.PrjPCB		Sheet 3 of 14



Title Phase_Measurements.SchDoc

Revision: Rev03 Design Engineer: D. Hufnagel

Project: UZ_D_Inverter.PrjPCB

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Date: 17.04.2023

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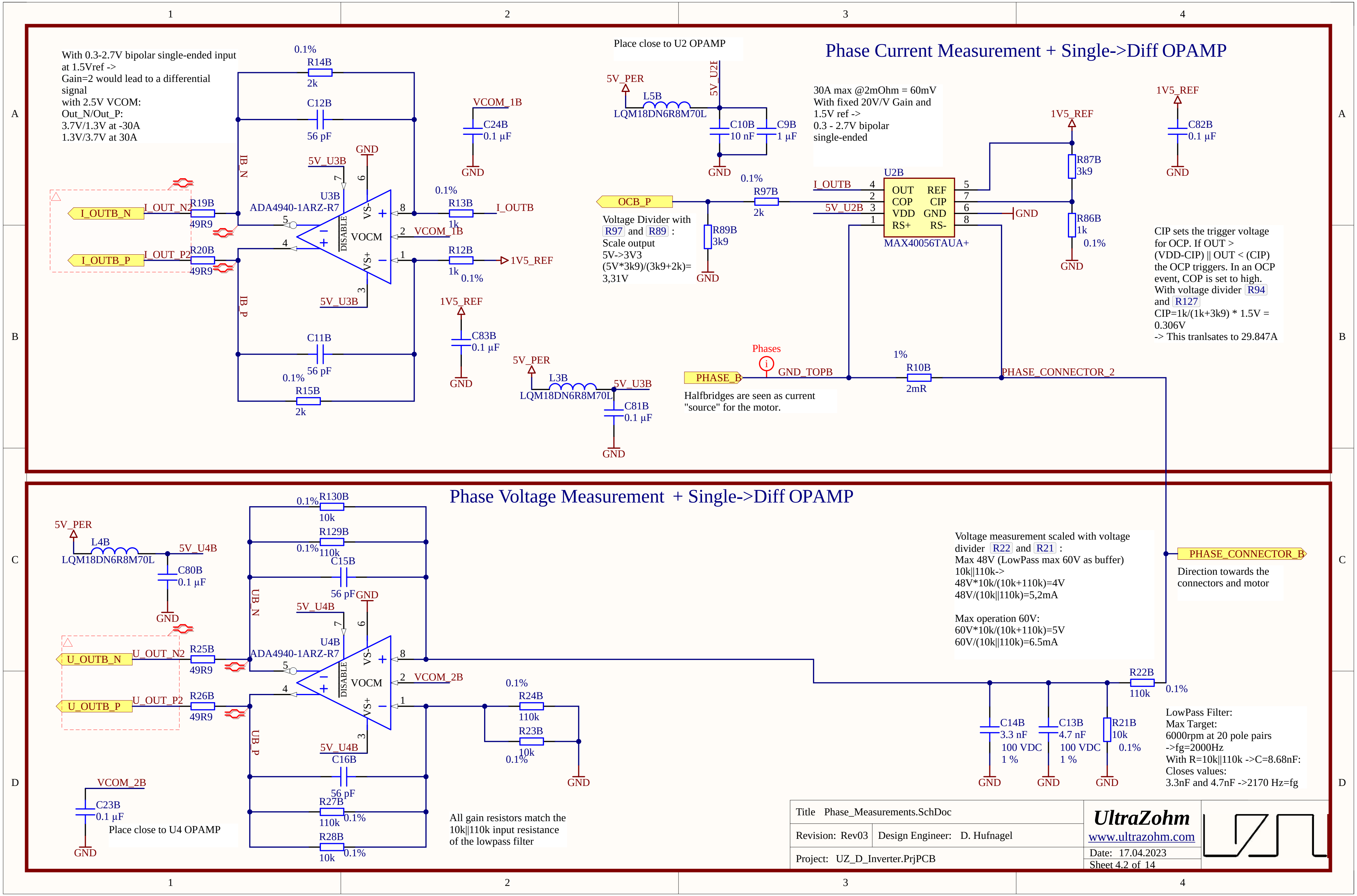
Phase Current Measurement + Single->Diff OPAMP
With 0.3-2.7V bipolar single-ended input at 1.5Vref -> Gain=2 would lead to a differential signal with 2.5V VCOM: Out_N/Out_P: 3.7V/1.3V at -30A 1.3V/3.7V at 30A
R14B 2k 0.1%
C12B 56 pF
VCOM_1B
C24B 0.1 µF
GND
5V_U3B
U3B ADA4940-1ARZ-R7
VOCM
DISABLE
I_OUTB_N
I_OUT_N2
R19B 49R9
I_OUTB_P
I_OUT_P2
R20B 49R9
IB_N
IB_P
R13B 1k 0.1%
I_OUTB
R12B 1k 0.1%
1V5_REF
C83B 0.1 µF
C11B 56 pF
R15B 2k 0.1%
Place close to U2 OPAMP
5V_PER
L5B LQM18DN6R8M70L
5V_U2B
C10B 10 nF
C9B 1 µF
30A max @2mOhm = 60mV With fixed 20V/V Gain and 1.5V ref -> 0.3 - 2.7V bipolar single-ended
OCB_P
R97B 2k 0.1%
R89B 3k9
Voltage Divider with R97 and R89 : Scale output 5V->3V3 (5V*3k9)/(3k9+2k)= 3,31V
U2B MAX40056TAUA+
OUT COP VDD RS+ REF CIP GND RS-
R87B 3k9
R86B 1k 0.1%
C82B 0.1 µF
CIP sets the trigger voltage for OCP. If OUT > (VDD-CIP) || OUT < (CIP) the OCP triggers. In an OCP event, COP is set to high. With voltage divider R94 and R127 CIP=1k/(1k+3k9) * 1.5V = 0.306V -> This tranlsates to 29.847A
L3B LQM18DN6R8M70L
5V_U3B
C81B 0.1 µF
Phases
PHASE_B
GND_TOPB
R10B 2mR 1%
PHASE_CONNECTOR_2
Halfbridges are seen as current "source" for the motor.
Phase Voltage Measurement + Single->Diff OPAMP
L4B LQM18DN6R8M70L
5V_U4B
C80B 0.1 µF
R130B 10k 0.1%
R129B 110k 0.1%
C15B 56 pF
U4B ADA4940-1ARZ-R7
U_OUTB_N
U_OUT_N2
R25B 49R9
U_OUTB_P
U_OUT_P2
R26B 49R9
UB_N
UB_P
VCOM_2B
R24B 110k 0.1%
R23B 10k 0.1%
C16B 56 pF
R27B 110k 0.1%
R28B 10k 0.1%
C23B 0.1 µF
Place close to U4 OPAMP
All gain resistors match the 10k||110k input resistance of the lowpass filter
Voltage measurement scaled with voltage divider R22 and R21 : Max 48V (LowPass max 60V as buffer) 10k||110k-> 48V*10k/(10k+110k)=4V 48V/(10k||110k)=5,2mA
Max operation 60V: 60V*10k/(10k+110k)=5V 60V/(10k||110k)=6.5mA
PHASE_CONNECTOR_B
Direction towards the connectors and motor
R22B 110k 0.1%
C14B 3.3 nF 100 VDC 1 %
C13B 4.7 nF 100 VDC 1 %
R21B 10k 0.1%
LowPass Filter: Max Target: 6000rpm at 20 pole pairs ->fg=2000Hz With R=10k||110k ->C=8.68nF: Closes values: 3.3nF and 4.7nF ->2170 Hz=fg
Title Phase_Measurements.SchDoc
Revision: Rev03
Design Engineer: D. Hufnagel
Project: UZ_D_Inverter.PrjPCB
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Date: 17.04.2023
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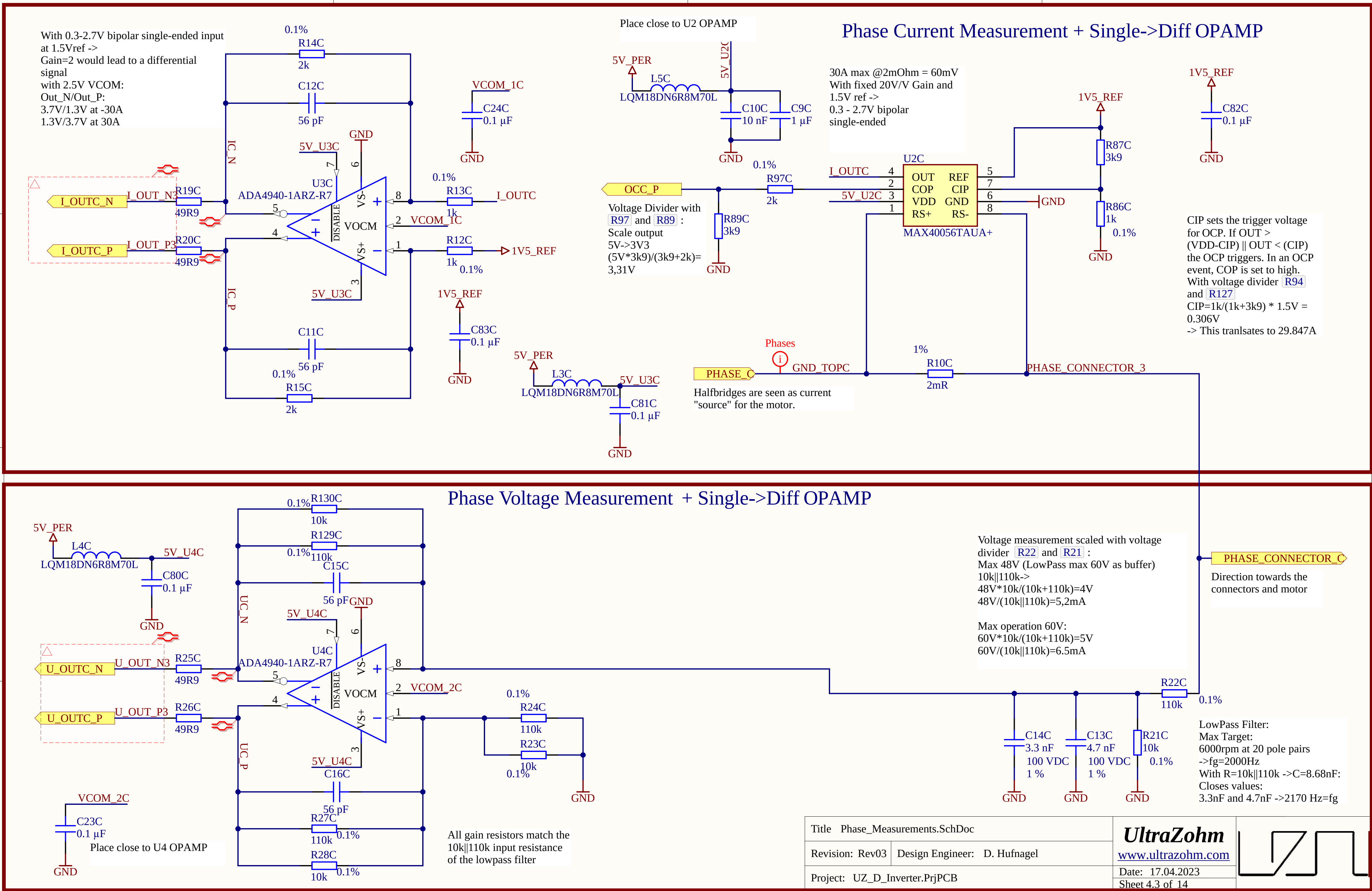

Phase Current Measurement + Single->Diff OPAMP
With 0.3-2.7V bipolar single-ended input at 1.5Vref -> Gain=2 would lead to a differential signal with 2.5V VCOM: Out_N/Out_P: 3.7V/1.3V at -30A 1.3V/3.7V at 30A
0.1% R14C 2k
C12C 56 pF
IC_N
IC_P
5V_U3C
GND
VCOM_1C
C24C 0.1 µF
GND
I_OUTC_N
I_OUT_N3
R19C 49R9
I_OUTC_P
I_OUT_P3
R20C 49R9
U3C ADA4940-1ARZ-R7
DISABLE
VOCM
VS-
VS+
0.1% R13C 1k I_OUTC
2 VCOM_1C
R12C 1k 0.1%
1V5_REF
1V5_REF
C83C 0.1 µF
GND
C11C 56 pF
0.1% R15C 2k
Place close to U2 OPAMP
5V_PER
L5C LQM18DN6R8M70L
5V_U2C
C10C 10 nF
C9C 1 µF
GND
30A max @2mOhm = 60mV With fixed 20V/V Gain and 1.5V ref -> 0.3 - 2.7V bipolar single-ended
1V5_REF
1V5_REF
C82C 0.1 µF
GND
OCC_P
0.1% R97C 2k
R89C 3k9
GND
Voltage Divider with R97 and R89 : Scale output 5V->3V3 (5V*3k9)/(3k9+2k)= 3,31V
U2C
I_OUTC
5V_U2C
OUT REF COP CIP VDD GND RS+ RS-
MAX40056TAUA+
GND
R87C 3k9
R86C 1k
0.1%
GND
CIP sets the trigger voltage for OCP. If OUT > (VDD-CIP) || OUT < (CIP) the OCP triggers. In an OCP event, COP is set to high. With voltage divider R94 and R127 CIP=1k/(1k+3k9) * 1.5V = 0.306V -> This tranlsates to 29.847A
Phases
PHASE_C
GND_TOPC
1% R10C 2mR
PHASE_CONNECTOR_3
Halfbridges are seen as current "source" for the motor.
5V_PER
L3C LQM18DN6R8M70L
5V_U3C
C81C 0.1 µF
GND
Phase Voltage Measurement + Single->Diff OPAMP
5V_PER
L4C LQM18DN6R8M70L
5V_U4C
C80C 0.1 µF
GND
0.1% R130C 10k
R129C 0.1% 110k
C15C 56 pF
GND
UC_N
UC_P
5V_U4C
U4C ADA4940-1ARZ-R7
U_OUTC_N
U_OUT_N3
R25C 49R9
U_OUTC_P
U_OUT_P3
R26C 49R9
2 VCOM_2C
0.1% R24C 110k
R23C 10k 0.1%
GND
5V_U4C
C16C 56 pF
R27C 110k 0.1%
R28C 10k 0.1%
VCOM_2C
C23C 0.1 µF
GND
Place close to U4 OPAMP
All gain resistors match the 10k||110k input resistance of the lowpass filter
Voltage measurement scaled with voltage divider R22 and R21 : Max 48V (LowPass max 60V as buffer) 10k||110k-> 48V*10k/(10k+110k)=4V 48V/(10k||110k)=5,2mA
Max operation 60V: 60V*10k/(10k+110k)=5V 60V/(10k||110k)=6.5mA
PHASE_CONNECTOR_C
Direction towards the connectors and motor
R22C 110k 0.1%
C14C 3.3 nF 100 VDC 1 %
C13C 4.7 nF 100 VDC 1 %
R21C 10k 0.1%
GND
LowPass Filter: Max Target: 6000rpm at 20 pole pairs ->fg=2000Hz With R=10k||110k ->C=8.68nF: Closes values: 3.3nF and 4.7nF ->2170 Hz=fg
Title Phase_Measurements.SchDoc
Revision: Rev03
Design Engineer: D. Hufnagel
Project: UZ_D_Inverter.PrjPCB
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Date: 17.04.2023
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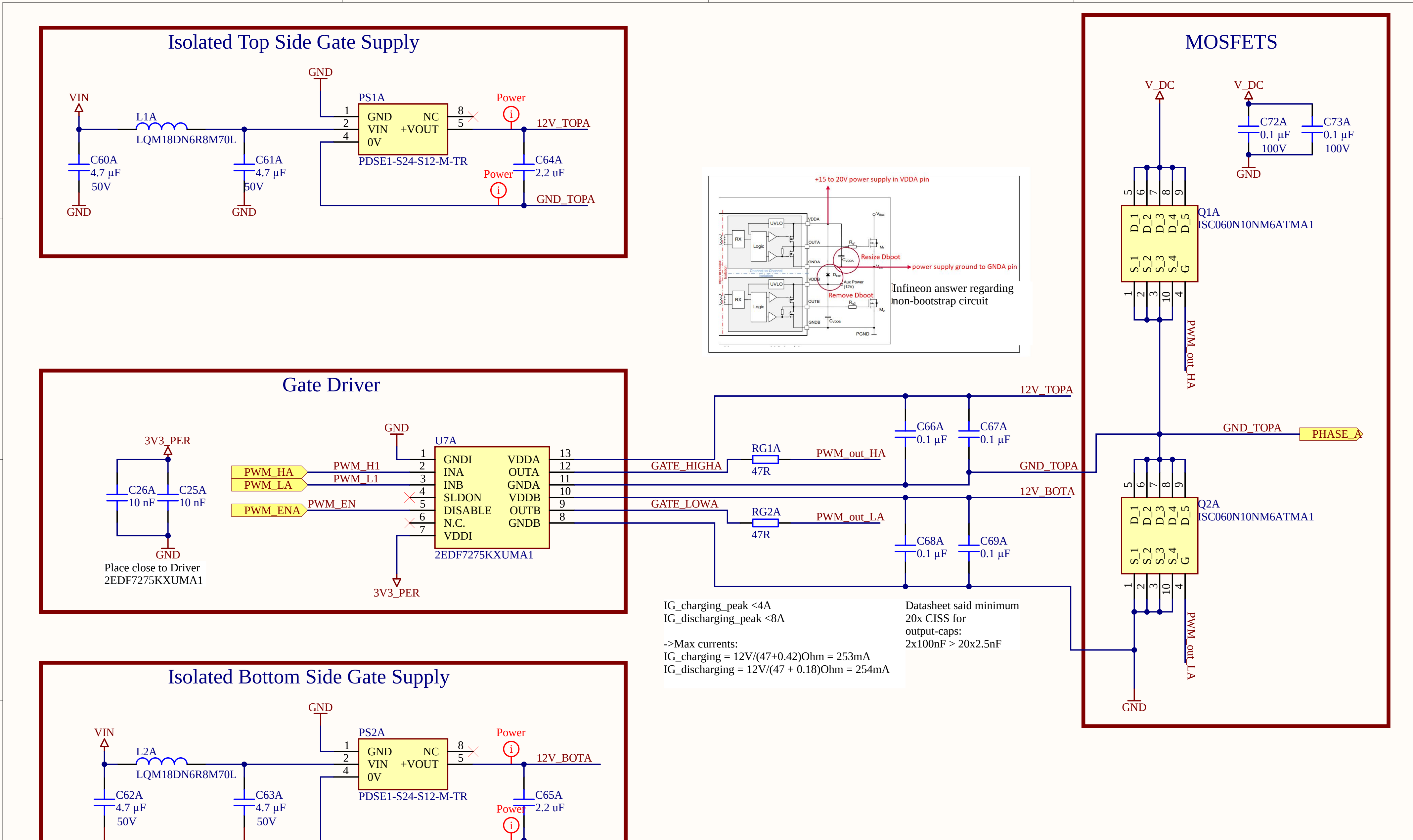
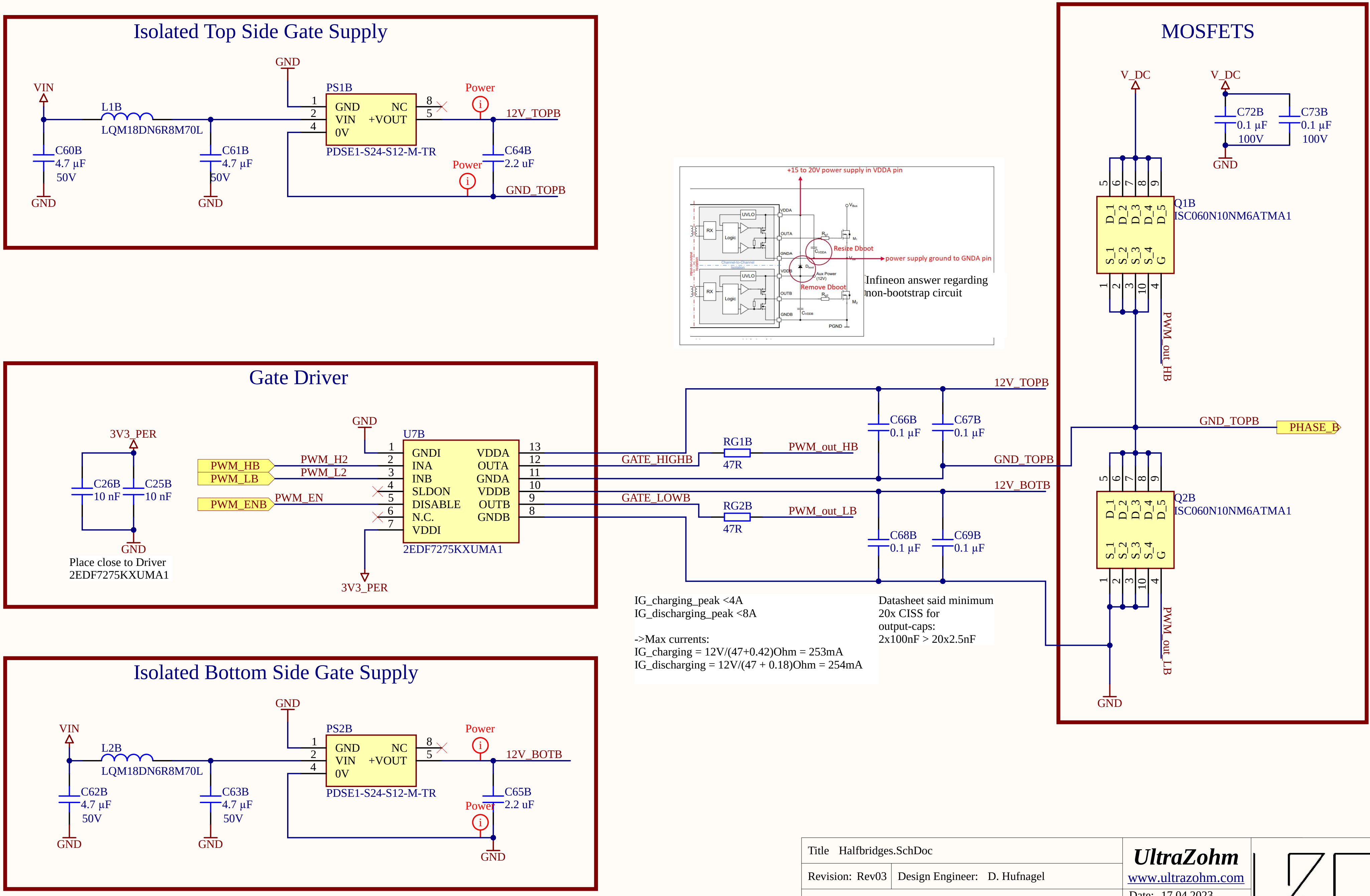


Title Halfbridges.SchDoc	UltraZohm
Revision: Rev03 Design Engineer: D. Hufnagel	www.ultrazohm.com
Project: UZ_D_Inverter.PrjPCB	Date: 17.04.2023 Sheet 5.1 of 14



Title Halfbridges.SchDoc

Revision: Rev03 | Design Engineer: D. Hufnagel

Project: UZ_D_Inverter.PrjPCB

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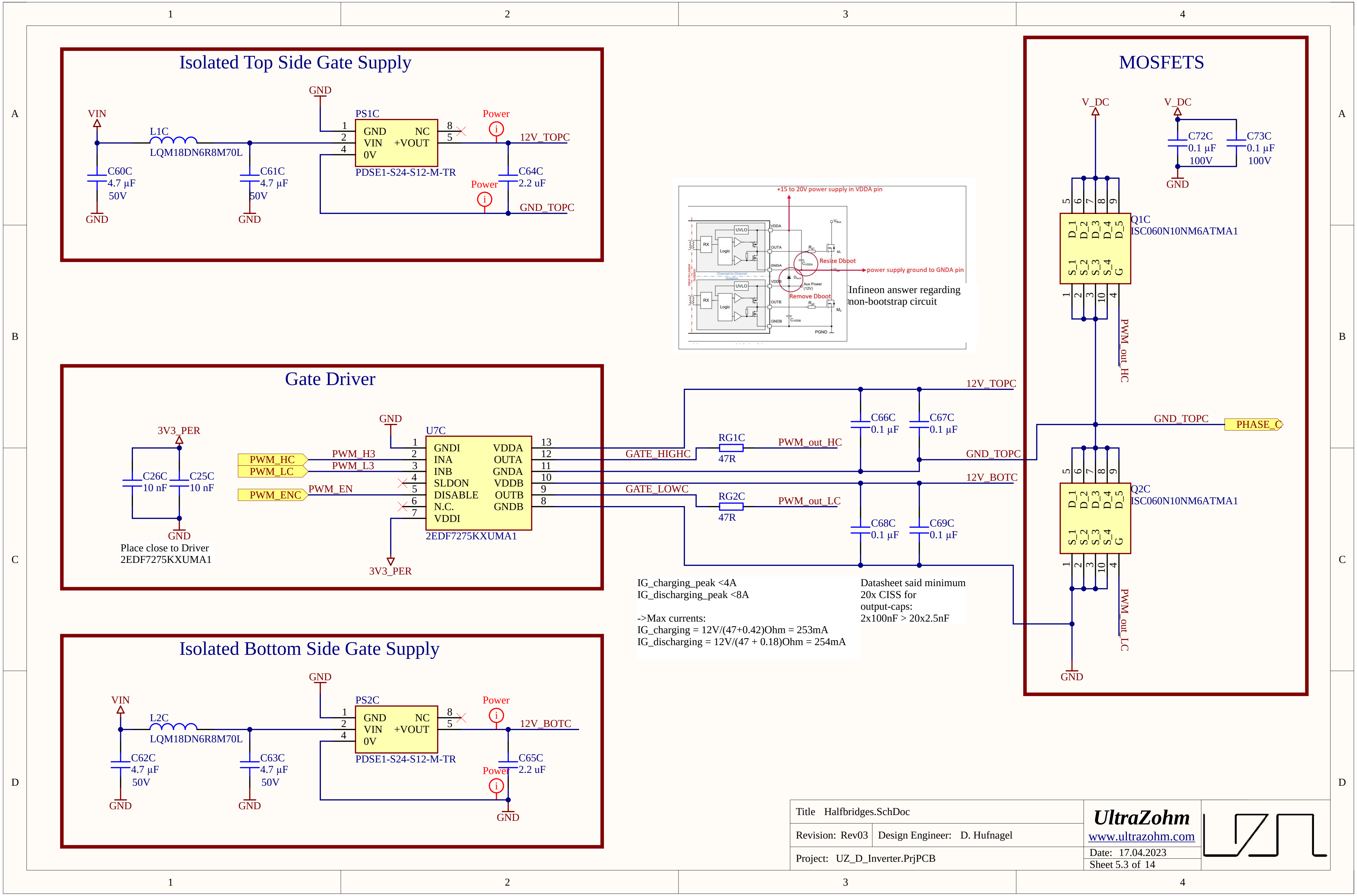
Isolated Top Side Gate Supply
VIN
L1C
LQM18DN6R8M70L
C60C
4.7 µF
50V
C61C
4.7 µF
50V
GND
PS1C
GND NC
VIN +VOUT
0V
PDSE1-S24-S12-M-TR
Power
12V_TOPC
C64C
2.2 uF
GND_TOPC
Gate Driver
3V3_PER
C26C
10 nF
C25C
10 nF
Place close to Driver
2EDF7275KXUMA1
PWM_HC
PWM_LC
PWM_ENC
PWM_H3
PWM_L3
PWM_EN
U7C
GNDI
INA
INB
SLDON
DISABLE
N.C.
VDDI
VDDA
OUTA
GNDA
VDDB
OUTB
GNDB
2EDF7275KXUMA1
GATE_HIGHC
GATE_LOWC
RG1C
47R
RG2C
47R
PWM_out_HC
PWM_out_LC
C66C
0.1 µF
C67C
0.1 µF
C68C
0.1 µF
C69C
0.1 µF
12V_TOPC
GND_TOPC
12V_BOTC
+15 to 20V power supply in VDDA pin
Resize Dboot
power supply ground to GNDA pin
Remove Dboot
Infineon answer regarding
non-bootstrap circuit
IG_charging_peak <4A
IG_discharging_peak <8A
->Max currents:
IG_charging = 12V/(47+0.42)Ohm = 253mA
IG_discharging = 12V/(47 + 0.18)Ohm = 254mA
Datasheet said minimum
20x CISS for
output-caps:
2x100nF > 20x2.5nF
MOSFETS
V_DC
C72C
0.1 µF
100V
C73C
0.1 µF
100V
Q1C
ISC060N10NM6ATMA1
Q2C
ISC060N10NM6ATMA1
D_1 D_2 D_3 D_4 D_5
S_1 S_2 S_3 S_4 G
GND_TOPC
PHASE_C
Isolated Bottom Side Gate Supply
VIN
L2C
LQM18DN6R8M70L
C62C
4.7 µF
50V
C63C
4.7 µF
50V
PS2C
PDSE1-S24-S12-M-TR
12V_BOTC
C65C
2.2 uF
Title Halfbridges.SchDoc
Revision: Rev03
Design Engineer: D. Hufnagel
Project: UZ_D_Inverter.PrjPCB
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Date: 17.04.2023
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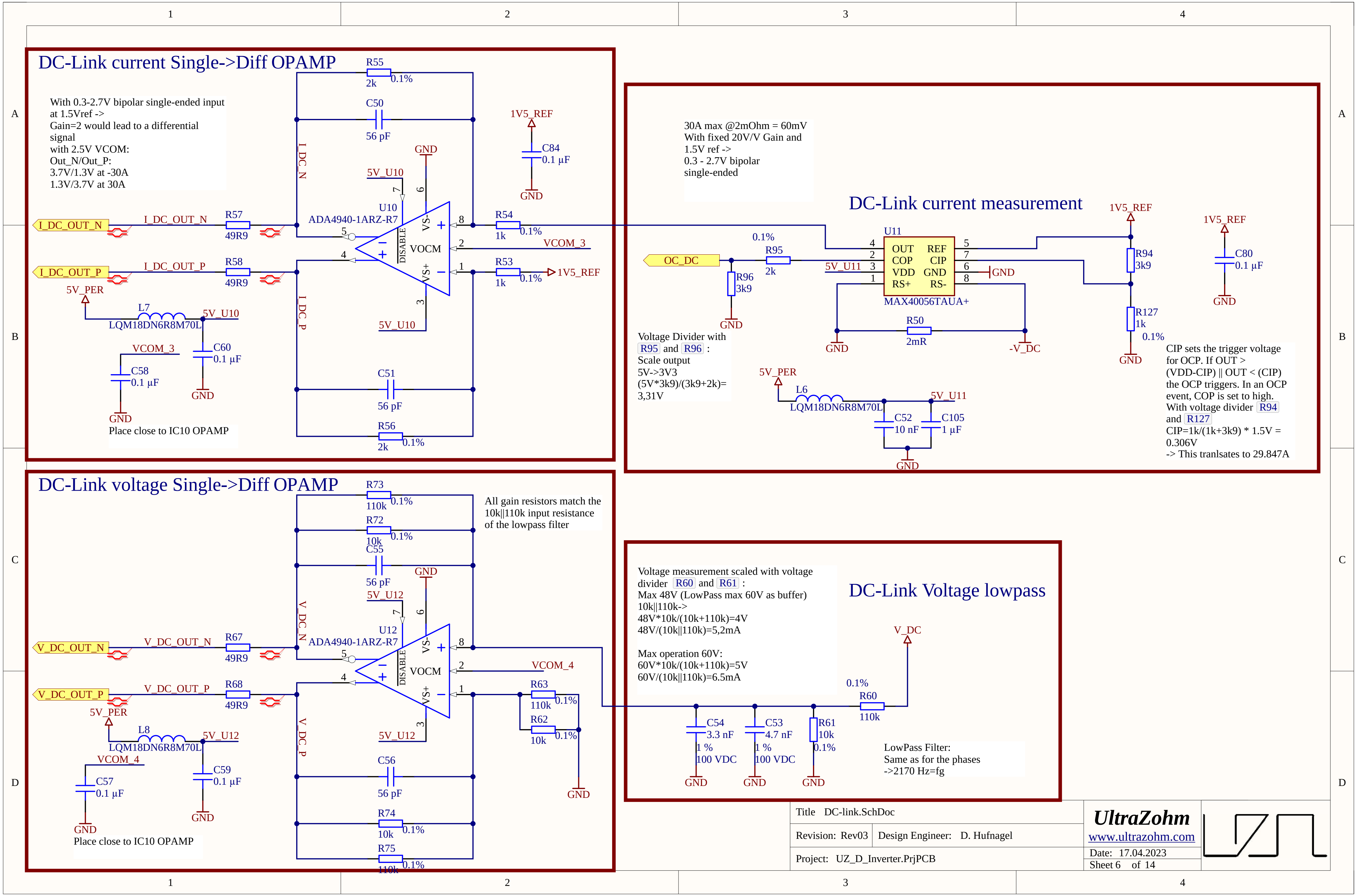

DC-Link current Single->Diff OPAMP
With 0.3-2.7V bipolar single-ended input at 1.5Vref -> Gain=2 would lead to a differential signal with 2.5V VCOM: Out_N/Out_P: 3.7V/1.3V at -30A 1.3V/3.7V at 30A
Place close to IC10 OPAMP
U10 ADA4940-1ARZ-R7
DC-Link current measurement
30A max @2mOhm = 60mV With fixed 20V/V Gain and 1.5V ref -> 0.3 - 2.7V bipolar single-ended
U11 MAX40056TAUA+
Voltage Divider with R95 and R96 : Scale output 5V->3V3 (5V*3k9)/(3k9+2k)= 3,31V
CIP sets the trigger voltage for OCP. If OUT > (VDD-CIP) || OUT < (CIP) the OCP triggers. In an OCP event, COP is set to high. With voltage divider R94 and R127 CIP=1k/(1k+3k9) * 1.5V = 0.306V -> This tranlsates to 29.847A
DC-Link voltage Single->Diff OPAMP
All gain resistors match the 10k||110k input resistance of the lowpass filter
U12 ADA4940-1ARZ-R7
Place close to IC10 OPAMP
DC-Link Voltage lowpass
Voltage measurement scaled with voltage divider R60 and R61 : Max 48V (LowPass max 60V as buffer) 10k||110k-> 48V*10k/(10k+110k)=4V 48V/(10k||110k)=5,2mA
Max operation 60V: 60V*10k/(10k+110k)=5V 60V/(10k||110k)=6.5mA
LowPass Filter: Same as for the phases ->2170 Hz=fg
Title DC-link.SchDoc
Revision: Rev03 Design Engineer: D. Hufnagel
Project: UZ_D_Inverter.PrjPCB
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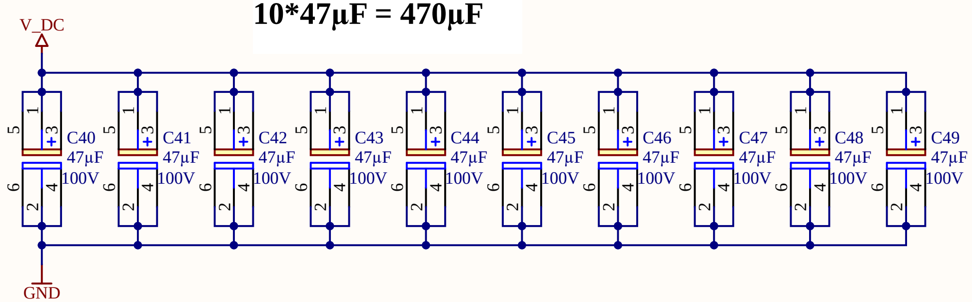
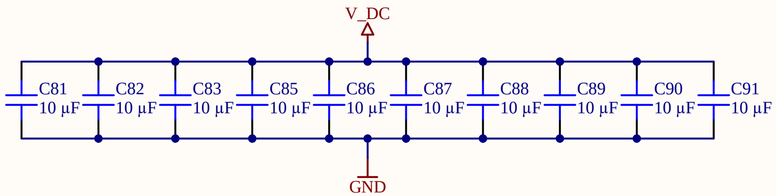
10*47µF = 470µF





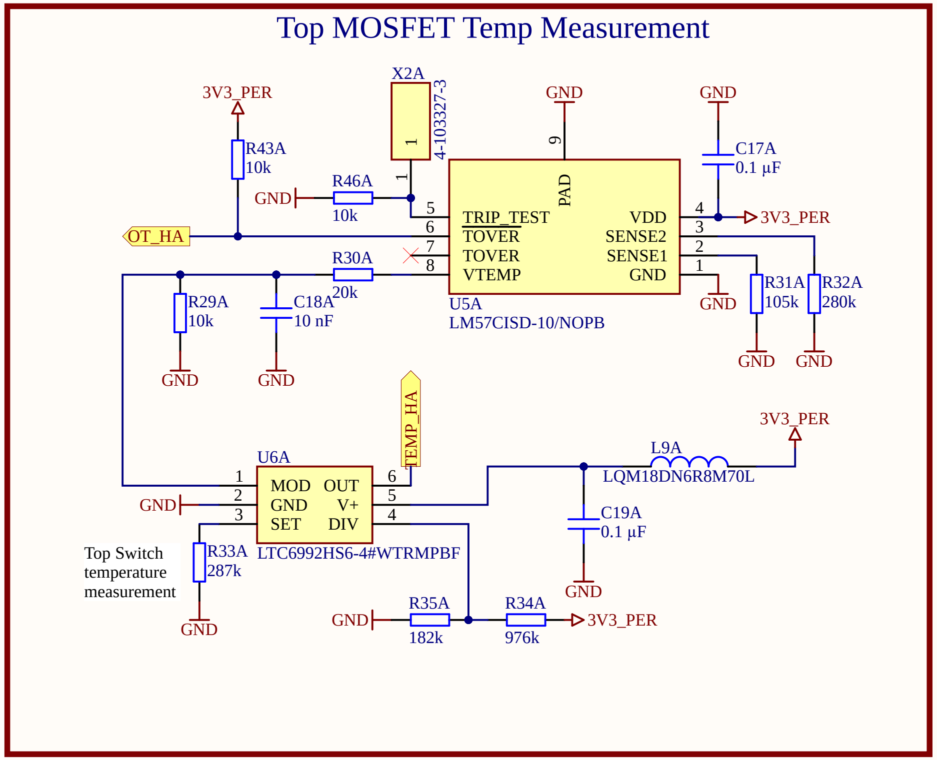
Additional MLCC caps with low internal R
10*10µF = 100µF

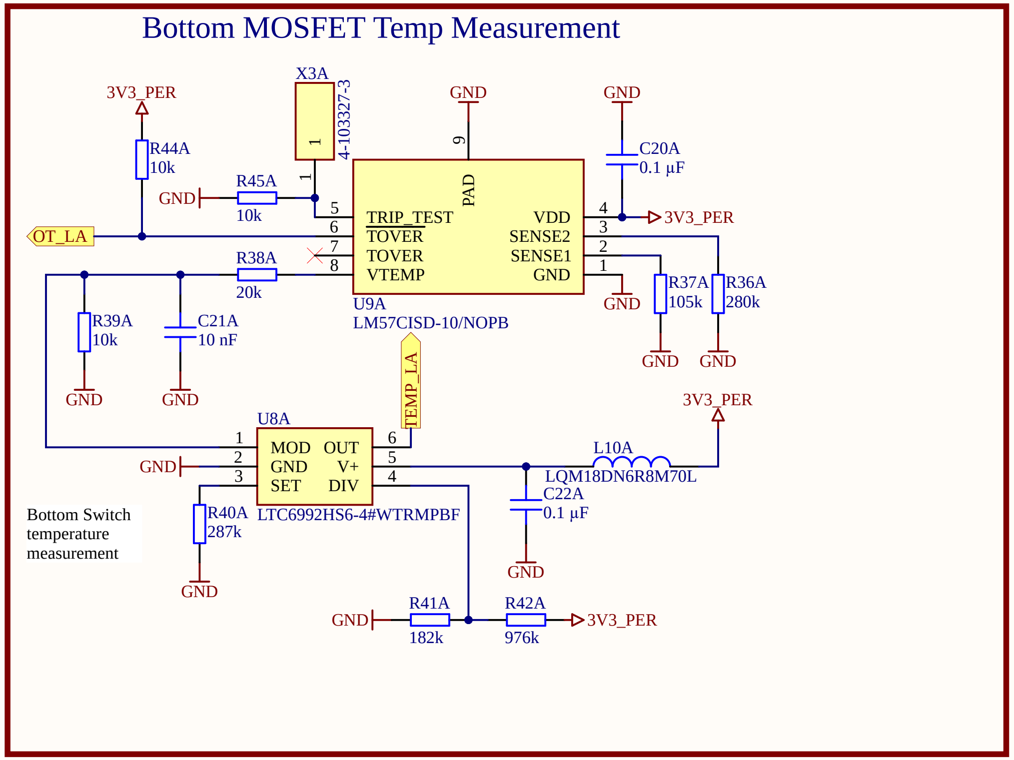
Total DC-Link capacitance
470µF + 100µF = 570 µF

Title DC-Link-Caps.SchDoc		*UltraZohm* www.ultrazohm.com
Revision: Rev03	Design Engineer: D. Hufnagel	
Project: UZ_D_Inverter.PrjPCB		Date: 17.04.2023 Sheet 7 of 14

1 2 3 4

A





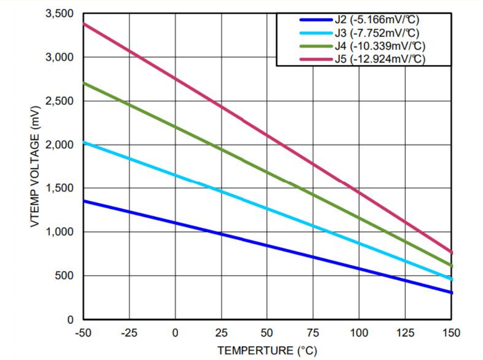
B

C

R_SENSE1 (R31 / R37) = 105k
R_SENSE2 (R32 / R36) = 280k
is Gain
->J4 ->Leads to:
1.11V@105°C
2.00V@20°C

VTEMP has voltage divider with 20k/10k:
MOD input voltages->
1.11V * 10/(10+20) = 0.37V@105°C
2.00V * 10/(10+20) = 0.67V@20°C

MOD input should be between 0.14V and 0.9V

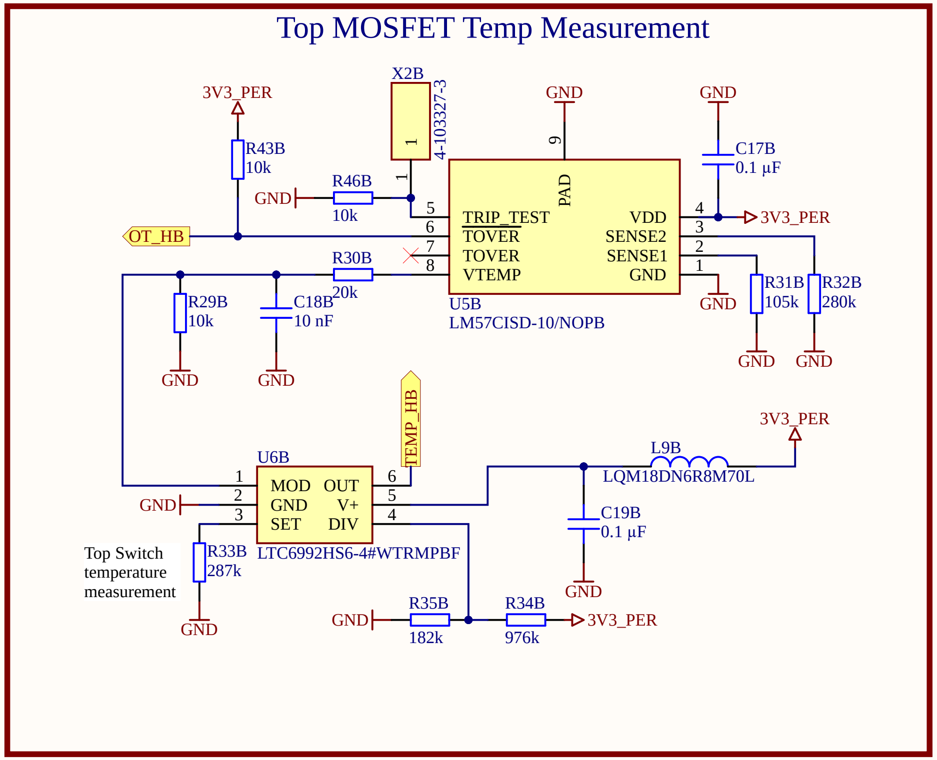


D

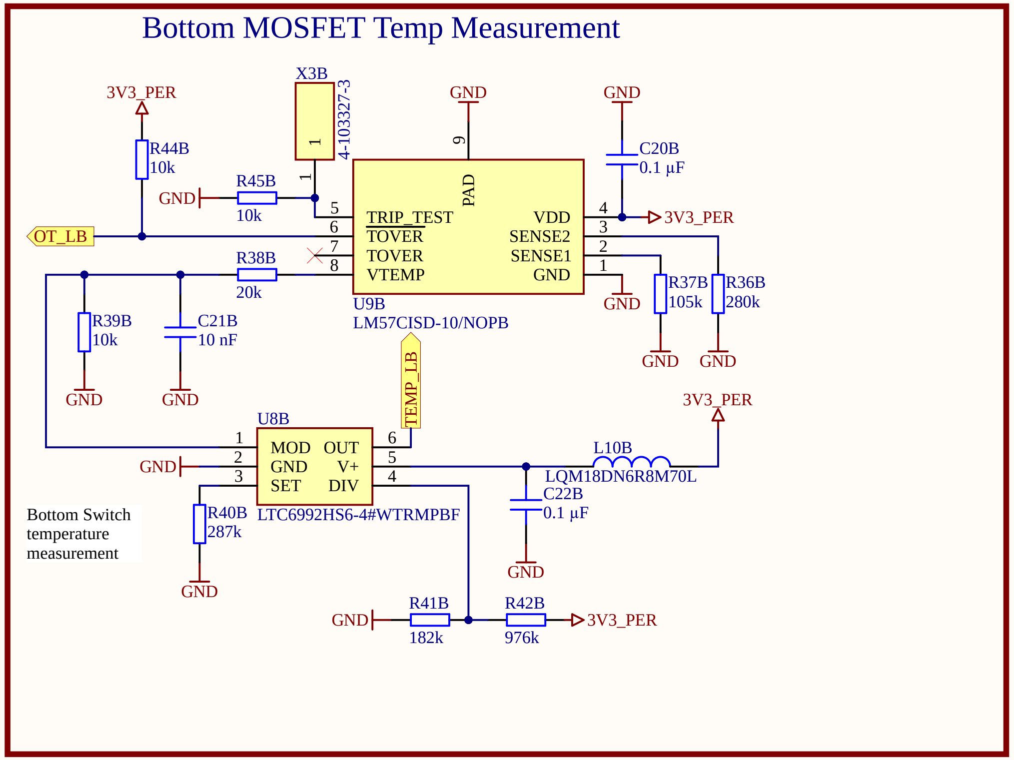
Title Temp_Measurements.SchDoc		**UltraZohm** www.ultrazohm.com
Revision: Rev03	Design Engineer: D. Hufnagel	Date: 17.04.2023
Project: UZ_D_Inverter.PrjPCB		Sheet 8.1 of 14

1 2 3 4

Top MOSFET Temp Measurement



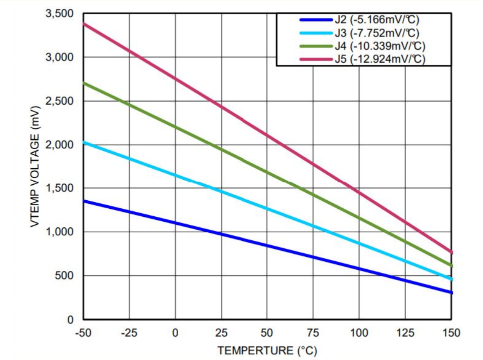
Bottom MOSFET Temp Measurement



R_SENSE1 (R31 / R37) = 105k
R_SENSE2 (R32 / R36) = 280k
is Gain
->J4 ->Leads to:
1.11V@105°C
2.00V@20°C

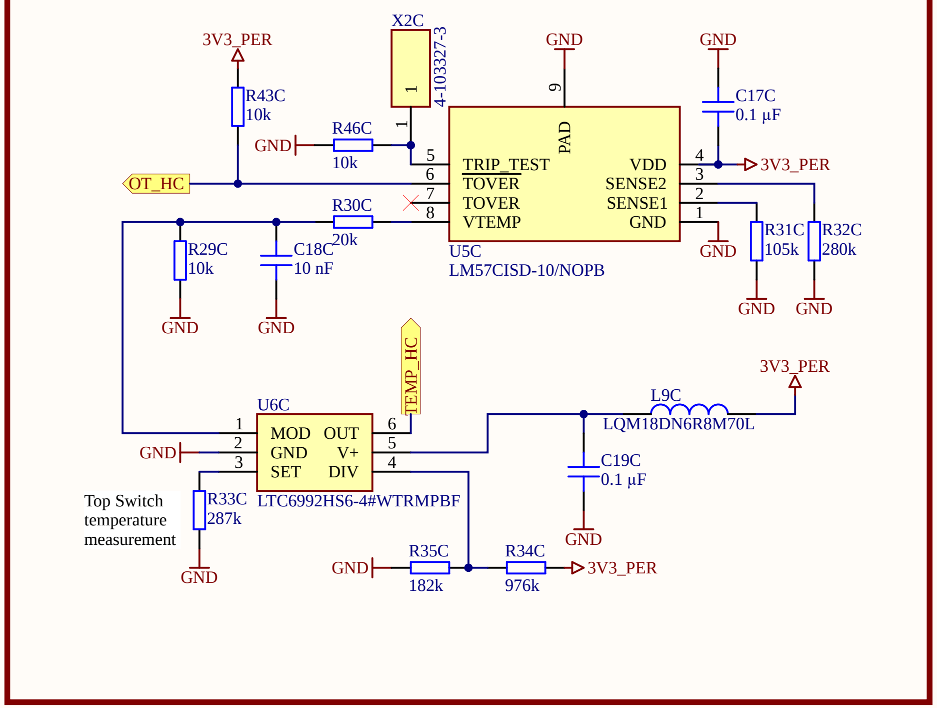
VTEMP has voltage divider with 20k/10k:
MOD input voltages->
1.11V * 10/(10+20) = 0.37V@105°C
2.00V * 10/(10+20) = 0.67V@20°C

MOD input should be between 0.14V and 0.9V

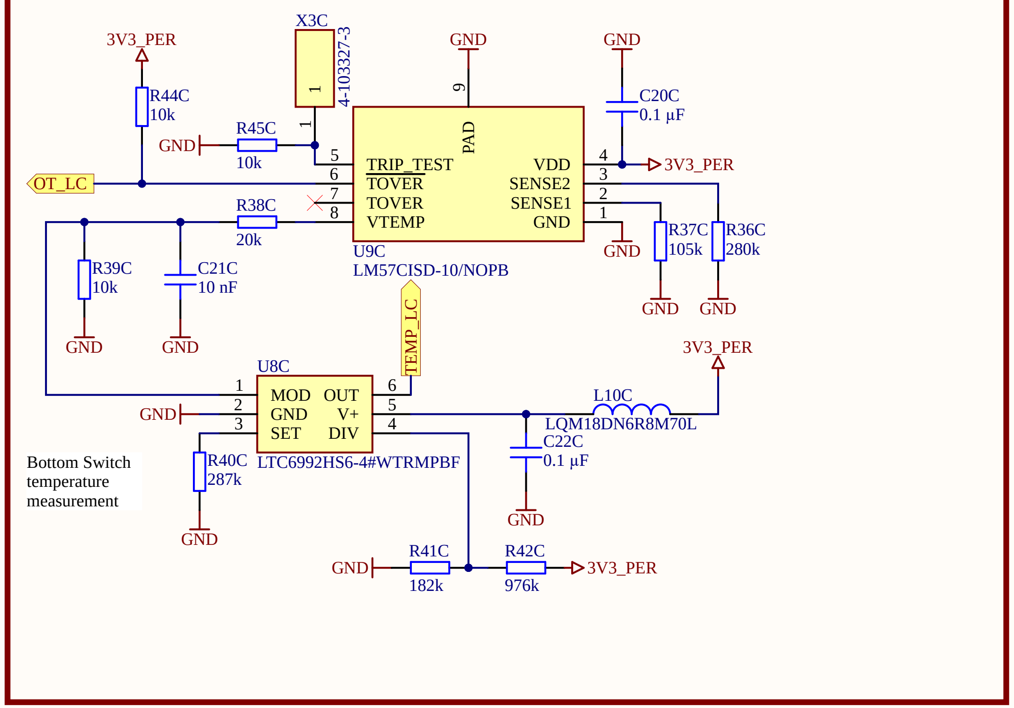


Title Temp_Measurements.SchDoc		UltraZohm www.ultrazohm.com
Revision: Rev03	Design Engineer: D. Hufnagel	
Project: UZ_D_Inverter.PrjPCB		Date: 17.04.2023 Sheet 8.2 of 14

Top MOSFET Temp Measurement



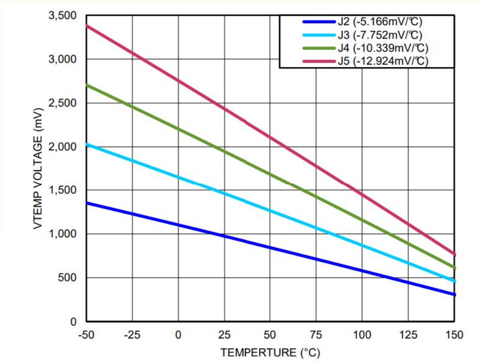
Bottom MOSFET Temp Measurement



R_SENSE1 (R31 / R37) = 105k
R_SENSE2 (R32 / R36) = 280k
is Gain
->J4 ->Leads to:
1.11V@105°C
2.00V@20°C

VTEMP has voltage divider with 20k/10k:
MOD input voltages->
1.11V * 10/(10+20) = 0.37V@105°C
2.00V * 10/(10+20) = 0.67V@20°C

MOD input should be between 0.14V and 0.9V



Title	Temp_Measurements.SchDoc
Revision: Rev03	Design Engineer: D. Hufnagel
Project: UZ_D_Inverter.PrjPCB	

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Date: 17.04.2023
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